

QUANTUM COMPUTING • LEARN BY EXPERIMENT

Read Your Results & Use H Twice

Learn how to interpret measurement results and discover why applying H twice takes a qubit back to where it started.

15 Read Your Results

We have learned how to build a circuit and predict what should happen. But when we actually run a quantum circuit, we do not usually get to see the quantum state directly.

Instead, we **measure the qubit** and record the result. When we repeat the circuit many times, we can look at the collection of results and understand the probabilities experimentally.

Key idea:

A single measurement gives one result. Many measurements help us see the probability distribution predicted by the quantum state.

15.1 One measurement gives one answer

Consider the circuit:



After applying H:

$$|0\rangle \rightarrow (|0\rangle + |1\rangle)/\sqrt{2}$$

Therefore:

$$P(0) = 50\% \quad P(1) = 50\%$$

If you measure it once, however, you do not receive “50% 0 and 50% 1.”

You receive either: **0** or **1**.

One-shot measurement

Possible result:



The probability tells us how likely each result is before the measurement.

15.2 Why run the circuit many times?

Because one measurement can be random, we usually repeat the same circuit many times.
Each individual measurement gives one result, but the overall collection of results reveals the underlying probabilities.

More measurements → clearer picture of the probabilities

15.3 Example: 10 measurements

Suppose we run the H circuit 10 times and obtain:

0, 1, 1, 0, 1, 0, 1, 1, 0, 1

Count how many times each result appeared.

Result	Number of times	Observed frequency
0	4	4/10 = 40%
1	6	6/10 = 60%

This does not contradict the 50–50 prediction. Ten measurements are simply not enough to guarantee exactly five of each.

15.4 What happens with more measurements?

Suppose we increase the number of measurements. The observed percentages should generally become closer to the expected probabilities.

0	48%
1	52%

With 100 measurements, we might see something like 48 zeros and 52 ones.

With 10,000 measurements, the percentages will typically be much closer to 50% and 50%.

15.5 Counts vs probabilities

Term	Meaning
Probability	The theoretical chance of obtaining a particular outcome.
Count	How many times that outcome actually appeared.
Frequency	The fraction or percentage of measurements producing that outcome.

15.6 From counts to probabilities

If a circuit is run **N** times and the result 0 appears **n₀** times, the observed frequency of 0 is:

$$\text{Observed frequency of 0} = n_0 / N$$

Similarly, if 1 appears **n₁** times:

$$\text{Observed frequency of 1} = n_1 / N$$

Since every measurement gives either 0 or 1:

$$n_0 + n_1 = N$$

15.7 Example with 1,000 shots

Suppose the circuit:

$|0\rangle$

H

M

is run 1,000 times.

Imagine the results are:

Result 0

497 times

Result 1

503 times

The observed frequencies are:

$$\hat{P}(0) = 497/1000 = 49.7\%$$

$$\hat{P}(1) = 503/1000 = 50.3\%$$

These values are very close to the theoretical prediction of 50% and 50%.

15.8 The word “shots”

In quantum computing, a repeated execution of the same circuit is often called a **shot**.

Number of shots	What it means
1 shot	One measurement
100 shots	The circuit is measured 100 times
1,000 shots	The circuit is measured 1,000 times

15.9 Read the result and compare it with your prediction

Prediction

What probabilities did your quantum state predict?

Result

What percentages did your measurements actually produce?

Compare

Are the experimental frequencies reasonably close?

Remember: Experimental counts do not need to exactly equal theoretical probabilities. Randomness causes small fluctuations. What matters is that, with enough repetitions, the observed frequencies generally approach the predicted probabilities.

15.10 Quick check

Suppose a circuit predicts:

$$P(0) = 25\% \quad P(1) = 75\%$$

If you run it 100 times, which result would be reasonable?

25 zeros and 75 ones

50 zeros and 50 ones

75 zeros and 25 ones

Answer: About 25 zeros and 75 ones would be a reasonable result. The exact counts can vary from experiment to experiment.

Read your results like a scientist:

Predict the probabilities → run many shots → count the outcomes → calculate frequencies
→ compare them with your prediction.

16 What If We Use H Twice?

We already know that the Hadamard gate creates a superposition. So a natural question is:

What happens if we apply H twice?

If one H creates a 50–50 superposition, will a second H make the probabilities even more random?

Yes — still 50–50

No — something else happens

16.1 Start with $|0\rangle$

The initial state is:

$$|\psi\rangle = |0\rangle$$

Apply the first H gate.

 $|0\rangle$

H

M

We know:

$$H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$$

So after the first H, the qubit is in an equal superposition.

16.2 Add another H

Now replace the measurement with another H gate.



The circuit is:

$$|0\rangle \rightarrow H \rightarrow H \rightarrow M$$

16.3 The surprising result

The second H does not leave us with a 50–50 result. Instead, it brings the qubit back to its original state:

$$H(H|0\rangle) = |0\rangle$$

Therefore:

$$P(0) = 100\% \quad P(1) = 0\%$$

Important:

H does not simply “add randomness.” It changes amplitudes, including their relative signs. A second H can use those amplitudes to reconstruct the original state.

16.4 Let's calculate it step by step

Start with:

$$|0\rangle$$

After the first H:

$$|0\rangle \rightarrow (|0\rangle + |1\rangle)/\sqrt{2}$$

Now apply H to both parts:

$$H[(|0\rangle + |1\rangle)/\sqrt{2}] = [H|0\rangle + H|1\rangle]/\sqrt{2}$$

We know:

First transformation

$$H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$$

Second transformation

$$H|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$$

Substitute them:

$$1/\sqrt{2} [(|0\rangle + |1\rangle)/\sqrt{2} + (|0\rangle - |1\rangle)/\sqrt{2}]$$

Notice what happens to the $|1\rangle$ terms:

$$+|1\rangle - |1\rangle = 0$$

They cancel.

The $|0\rangle$ terms add together, leaving:

$$|0\rangle$$

16.5 This is interference

The cancellation above is an example of **quantum interference**.

Quantum amplitudes can combine with each other. Because amplitudes can be positive, negative, or complex, they can reinforce or cancel one another.

Constructive interference

Amplitudes reinforce each other.

$$+a + a = 2a$$

Destructive interference

Amplitudes cancel each other.

$$+a - a = 0$$

This is one of the most important ideas in quantum computing: **the amplitudes interfere before probabilities are measured.**

16.6 Why doesn't this happen with ordinary probabilities?

Suppose you had ordinary classical probabilities of 50% and 50%. You cannot simply attach a negative sign to a probability and make probabilities cancel.

Probabilities are always non-negative:

$$P \geq 0$$

Quantum amplitudes are different. They can have signs and phases, allowing interference.

Quantity	Can be negative?	Can interfere?
Probability	No	No
Amplitude	Yes	Yes

16.7 The mathematical shortcut

The fact that applying H twice returns the original state can be written very compactly as:

$$H^2 = I$$

Here, **I** is the identity operation. It means “do nothing” to the state.

$$H \times H = I$$

Therefore:

$$H(H|\psi\rangle) = |\psi\rangle$$

This is true for any single-qubit state $|\psi\rangle$, not just $|0\rangle$.

16.8 What if we start with $|1\rangle$?

Let's test the same idea starting from $|1\rangle$.

$|1\rangle$

H

H

M

First H:

$$H|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$$

Second H:

$$H(H|1\rangle) = |1\rangle$$

Prediction: Starting with $|1\rangle$, applying H twice returns the qubit to $|1\rangle$. Therefore, measurement gives 1 with probability 100%.

16.9 Compare one H with two H gates

Circuit	Final state	Measurement probabilities
$ 0\rangle \rightarrow M$	$ 0\rangle$	0 = 100%
$ 0\rangle \rightarrow H \rightarrow M$	$(0\rangle + 1\rangle)/\sqrt{2}$	0 = 50%, 1 = 50%
$ 0\rangle \rightarrow H \rightarrow H \rightarrow M$	$ 0\rangle$	0 = 100%

16.10 A useful mental picture

Before H

The qubit is definitely in $|0\rangle$.

$|0\rangle$

After H

The qubit has two amplitudes.

$+|0\rangle + |1\rangle$

After H again

The amplitudes recombine and return to $|0\rangle$.

$|0\rangle$

16.11 Quick prediction challenge

Predict before calculating

What happens in this circuit?

$|0\rangle$

H

H

X

M

Always 0

Always 1

50–50

Solution:

The two H gates cancel each other: $H \rightarrow H = I$. So the circuit becomes effectively: $|0\rangle \rightarrow X \rightarrow |1\rangle \rightarrow M$. Therefore, the answer is **always 1** in an ideal circuit.

16.12 What you should remember

- ✓ One H creates a superposition from $|0\rangle$ or $|1\rangle$.
- ✓ A second H does not simply create “more randomness.”
- ✓ The amplitudes produced by the first H can interfere when the second H is applied.
- ✓ The $|1\rangle$ contributions can cancel while the $|0\rangle$ contributions reinforce.
- ✓ Applying H twice gives the identity: $H^2 = I$.
- ✓ Therefore, H twice returns any single-qubit state to its original state.

The big idea:

A quantum circuit is not just manipulating probabilities. It manipulates amplitudes. Those

amplitudes can interfere with one another, which is why applying H twice can completely undo the effect of the first H.

Quantum Computing Notes • Predict → Build → Measure → Read → Understand